

Francesco Cavina

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Data and AI engineer with a BSc in Econometrics and Data Science from VU Amsterdam and incoming MSc studies in Econometrics and Operations Research and Business Analytics. Experience building privacy aware AI workflows, governed data interfaces, automated pipelines and full stack prototypes. Strong interest in agentic automation, internal process improvement and practical AI tools that help teams reduce repetitive work and make better operational decisions.

TECHNICAL SKILLS

- **AI and Automation:** LLM workflows, RAG, agentic workflows, OpenAI API, Hugging Face, ChromaDB, prompt design, AI assisted data discovery, process automation concepts
- **Data Engineering:** Python, SQL, PySpark, Databricks, Airflow, metadata driven ingestion, schema inference, data quality workflows
- **Low Code and Product Thinking:** Power Automate basics, Copilot Studio interest, workflow design, documentation, user focused prototypes, internal tooling
- **Web and APIs:** FastAPI, Next.js, React, TypeScript, JavaScript, REST APIs, Tailwind CSS
- **Infrastructure:** Git, Docker, Kubernetes, Terraform, Azure

EXPERIENCE

Data & AI Platform Engineer

Robodata
Netherlands

September 2025 - June 2026, Amsterdam,

- Built privacy aware AI and data platform components for enterprise clients, including metadata driven ingestion, schema inference and PII classification workflows.
- Developed automated data pipelines in Python, PySpark and SQL using Bronze, Silver and Gold architecture.
- Built local LLM workflows for data discovery, field classification and metadata generation, with emphasis on governed use of enterprise data.
- Created AI ready data interfaces that allow internal datasets to be queried safely by LLM based agents.
- Worked on self service data tooling so non specialist users could understand, operate and reuse technical data infrastructure.
- Documented platform components, workflows and design choices to support handover, adoption and future scaling.
- Used Docker, Kubernetes, Terraform, Azure, Airflow and SQLAlchemy across production platform work.

Operational Trader

QuantFi
Netherlands

October 2022 - May 2023, Amsterdam,

- Developed and tested ML based trading strategies in Python using Random Forest, XGBoost, neural networks and walk forward validation.
- Built data ingestion, retraining and daily forecast workflows using Python and SQL.
- Created performance dashboards and visualisations to evaluate model output and support daily decisions.

Teaching Assistant

Vrije Universiteit Amsterdam
Netherlands

June 2022 - November 2024, Amsterdam,

- Taught weekly Probability Theory tutorials to first year Econometrics students.
- Explained abstract mathematical concepts in a clear and practical way.
- Built a student portfolio system for first year students and represented the programme during Open Days.

EDUCATION

MSc in Business Analytics

Track Computational Intelligence • Vrije Universiteit Amsterdam
• Amsterdam, Netherlands • 2026-2029

MSc in Econometrics and Operations Research

Track Business Engineering and Data Driven Decision Making • Vrije Universiteit Amsterdam
• Amsterdam, Netherlands • 2026-2029

Bachelor of Science in Econometrics and Data Science

Vrije Universiteit Amsterdam • Amsterdam, Netherlands • 2021-2024 • 7.8/10 •
Thesis: Nonstandard Probabilities in an Internal Probability Space, grade 8.5/10

Minor in Pure Mathematics

University of Leeds
• Leeds, England • 2024 • 8.5/10

PROJECTS

Selected projects focused on AI workflows, automation, usable interfaces and data driven products.

- 1) [AI Powered Personal Website](https://github.com/FrancescoCavina02/My-Website) • <https://github.com/FrancescoCavina02/My-Website> • November 2025 - present
Built a full stack portfolio with a FastAPI backend that parses an Obsidian vault into a structured REST API for notes, quotes and chat functionality. Designed the project as a self service knowledge interface, combining backend automation, frontend usability and LLM based interaction. Used Next.js, TypeScript, Python and Tailwind CSS.
- 2) [RAG Knowledge Chatbot](https://github.com/FrancescoCavina02/Spiritual-chatbot) • <https://github.com/FrancescoCavina02/Spiritual-chatbot> • November 2025 - present
Built a retrieval augmented chatbot using semantic chunking, vector search, structured prompts and citation based responses. Focused on reliable information retrieval, traceable answers and reusable workflow design. Used Python, FastAPI, ChromaDB, sentence transformers and LLM APIs.
- 3) [Dodgeball Club Amsterdam](https://github.com/FrancescoCavina02/DCA-website) • <https://github.com/FrancescoCavina02/DCA-website> • June 2025 - present
Built and maintain a public website using HTML, CSS and JavaScript. Focused on clear navigation, simple content management and practical usability for club members.

Podcasts

[Podcast Link](#) - Created two independent interview podcasts with researchers in physics, cosmology, mathematics and philosophy.

RELEVANT STRENGTHS

- Practical AI implementation, not just experimentation
- Translating unclear workflows into technical prototypes
- Building governed data access for LLM based tools
- Explaining technical concepts clearly to non specialist users
- Comfortable working across data, backend, automation and user interface layers